FEASIBILITY OF STELLAR DETECTION VIA NEUTRINO RATES

REPORT

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Contents

1 Abstract **3**

2 Introduction **4**

3 Theoretical foundations **5**

3.1 Neutrino emissions for a 20 solar mass star 5

3.1.1 Thermal and WNI neutrino production of a neutrino-cooled star 7

3.1.2 Pair annihilation neutrino spectra 10

3.2 Neutrino Mixing, Oscillations and Hierarchies 12

3.2.1 Computing MSW Oscillation effect 13

3.3 Detectability 16

3.3.1 Neutrino-electron scattering 16

3.3.2 Inverse Beta Decay 18

3.3.3 Other methods of detection 19

3.4 Detectors 20

3.4.1 Water Cherenkov 20

3.4.2 Liquid Scintillator 20

3.4.3 Detectors considered 21

4 Methodology **23**

5 Results **25**

6 Conclusions **28**

7 Bibliography **29**

8 Apendix 1 **33**

9 Apendix 2 **34**

1. Abstract

For a **20 solar mass star** in its **neutrino-cooled stage**, the **rates** of all flavors of **pair-annihilation neutrinos** are calculated as a function of time. In the best case scenario, for a star situated at 0.2 kpc of distance, our results show that we can detect the star via neutrinos up to **6 months** before core collapse. Additionally, study of the pre-collapse star seems to be **impossible beyond 10 kpc**. When performing the calculations, the 2 Liquid Scintillator (LS) detectors **JUNO** and **KamLAND** and the Water Cherenkov (WC) detector **SK-GD** have been taken into consideration.

2. Introduction

Stars produce neutrinos in their **cores**, the detection of these neutrinos provides precise information on the conditions of the medium in which they were created (temperature, density, chemical composition...etc). They also serve as **chronometers** of stellar evolution; different phases of a star have a different neutrino signature. In addition, neutrinos interact with matter only through the **weak nuclear interaction**, and because of this, they escape more easily than photons from the interior of stars. This is especially noticeable in stars in more advanced phases of their life where the opacity of the core is very considerable.

In this sense, if we could detect the neutrinos produced by a star, they would provide us with **more information** than that of photons. However the study of stellar neutrinos has focused, essentially, on the study of supernovae such as **SN 1987A** or, on the study of neutrinos in an **immediate pre-supernova** context which are considerably more energetic, and hence, easier to detect than those which precede them. In this work, we aim to resolve what is the real possibility of studying, through pair-annihilation neutrinos, these **phases that precede the immediate pre-collapse one**.

Some studies have already been conducted in this regard. The work of Kai Zuber and Santiago Díaz [1] shows that it is **impossible** to study stars in their H burning phase at reasonably large distances. In this phase, for massive stars, neutrinos produced by the **CNO cycle** dominate, however, unfortunately they are **not energetic enough** to be detected with our current technology. With the main sequence out of the question, we are left with the more **advances stages** of evolution. In this work, we speculate that for the burning phases of **Carbon, Neon, Oxygen and Silicon**; neutrino **detection** might be **possible** at reasonably close distances. Andrezj Odrzywolek [30] refers to the stars in this regime as **neutrino-cooled stars**. During this neutrino-cooled stage, the main neutrino flux is the one produced by the pair annihilation process. For simplicity, it will be this kind of neutrinos that we will focus on on this work.

Thus, we will focus on the study of these neutrino-cooled stars by detecting neutrinos produced by pair annihilation processes. Specifically, we will study the case of a **20M0** star for **different distances**, 3 **different metallicities** ([Fe/H] = 0.5, 0, -1) **and for 3 different detectors**. These are the Liquid Scintillator detectors JUNO and KamLAND and the Gadolinium-doped Water Cherenkov detector SK-GD.

3. Theoretical foundations

3.1. Neutrino emissions for a 20 solar mass star

There are **3 different flavors of neutrinos**, electronic (v_e), mu (v_μ) and tau (v_τ). The processes from which they are emitted differ during the stellar evolution of a star. For massive stars [28], neutrino emission can be divided into **six different stages** that correspond to different phases of its evolution. An HR neutrino luminosity diagram can be seen in **Appendix 1**.

1. H burning

For a massive stars [7], such as a 20 solar mass star, the primary source of neutrinos during H burning is the **Carbon–Nitrogen–Oxygen (CNO) cycle**. Which is highly temperature-sensitive $\epsilon_{CNO} \propto T^{17}$ [35], whereas PP is negligible. As a whole, v_e neutrinos produced by the CNO cycle during H burning account for a total liberated energy of only one order of magnitude below the energy liberated during SN (but liberated at a very slow pace). As we have already mentioned, due to its low energy, they are **not detectable** [1]. They are not detectable even if integrated over the whole galaxy [28].

2. He burning

As shown in Appendix 1, neutrino luminosity decreases during He burning. Helium burning is dominated by strong reactions that do not require weak decays, so the **main burning** does **not produce** neutrinos during this burning stage. In the core, only one weak process, namely ^{18}F decay generates neutrinos in the 3α reaction [32]. Hence, only the **thermal neutrinos** produced by the plasmon decay process can be considered **somewhat relevant**. Consequently, the neutrino luminosity during the helium burning phase arises primarily from the burning of the hydrogen shell. Tens of keV at best. Therefore, neutrinos from the He burning phase must be also considered **undetectable**.

3. Neutrino-cooled stage

Onto the onset of C burning. When the carbon is ignited, the internal temperature of the massive star exceeds $8 \cdot 10^8$ K, leading to the domination of thermal processes, particularly **electron–positron pair annihilation**, over weak nuclear interaction pro-

cesses (WNI) [7]. At this stage, massive stars are often called **neutrino-cooled stars** [30]. Electron positron pair-annihilation leads to a strong neutrino emission, with number of neutrinos emitted proportional to T^8 and the neutrino luminosity significantly surpasses the photon one. D. Arnett [37] proposed to refer it to as neutrino-cooled star i.e gravitationally confined thermonuclear reactor where energy generation by nuclear burning is balanced by emission of neutrino-antineutrino pairs, mainly from the pair annihilation process. This **situation continues** till the onset of silicon burning. therefore encompassing carbon, neon, oxygen and silicon burning stages.

4. Neutronization phase

Neutrino luminosity produced by the nuclear reactions becomes **gradually** important during Si burning, yet, it is not dominant until silicon burning is over. During this time, the number of electrons in the star decreases. This reduction can be attributed to an increase in the **internal density and temperature**, which leads to an increase in the electron degeneracy pressure. As a result **neutronization** occurs and the mean energy of electrons becomes larger than the 'Q-value' of many heavy nuclei, increasing the rate of e capture processes in the stellar interior [29]. Once silicon burning ends, and the core temperature reaches $(5 \cdot 10^9\ K)$ **nuclear statistical equilibrium** (NSE) is reached, and the e capture process is significantly enhanced becoming **dominant**.

5. Collapse

During NSE, β decay and electron capture (EC) on heavy nuclei i.e weak interaction reaction neutrinos, dominate the production of neutrinos. NSE continues after the end of all nuclear burning processes, and on into the onset of the collapse. After that, we enter the core-collapse stage of neutrino emissions.

6. Early and late cooling of proto-NS

The collapse is followed by the early and late cooling of the Pre Neutron Star (PNS) and the Neutron Star (NS) that forms afterwards. The early cooling of the PNS (time scale of 100 seconds) transforms the PNS into a NS contracting the radius, and releasing in the process an enormous gravitational binding energy of the order of 10^{53} in the form of neutrinos of all flavors. However, this is not the end, the NS continues to cool during thousands and millions of years. In this phase, thermal and weak interaction neutrinos seem to be **indistinguishable** [32].

Therefore, from a **neutrino physics** point of view, it is convenient to divide the entire life of a star into:

Stage	$\langle L_\nu \rangle$ [erg/s]	$E_\nu^{\rm tot}$ [erg]	Time	$\langle \mathcal{E}_\nu \rangle$ [MeV]	Process	Flavor
1.	10^{36}	10^{52}	10^7 yrs	0.5-1.7	CNO	ν_e
2.	10^{31}	10^{49}	10^6 yrs	0.02	plasma	all
3.	10^{38}–10^{46}	10^{51}	10^4 yrs	0.5–1.5	pair	all
4.	10^{54}	10^{51}	10^{-2} sec	10	ϵ^-	ν_e
5.	10^{52}–10^{48}	10^{53}	~ 100 sec	10–40	ν transport	all
6.	$< 10^{48}$	$< 10^{51}$	10^4 yrs	1	URCA	$\nu_e, \bar{\nu}_e$

As we have already mentioned, out of these 6 stages we are interested in the neutrino-cooled stage. This is, **number 3** on the table above.

3.1.1. Thermal and WNI neutrino production of a neutrino-cooled star

During the neutrino-cooled stage, neutrinos are produced via thermal and weak nuclear interaction processes. We have already commented on the fact that thermal neutrino dominate over WNI ones, especially pair-annihilation neutrinos do. But we still have to elucidate all contributions to both modes of neutrino production.

Thermal neutrinos:

There are 4 main processes involved in their production. These are:

- Electron–positron pair annihilation: $e^+ + e^- \longrightarrow \nu_x + \overline{\nu}_x$
- Plasmon decay: $\gamma^* \longrightarrow \nu_x + \overline{\nu}_x$
- Photo neutrino: $e^- + \gamma \longrightarrow \nu + \overline{\nu}_x$
- Bremsstrahlung: $e^- + (Z, A) \longrightarrow e^- + (Z, A) + \nu_x + \overline{\nu}_x$

These processes produce **all flavours of neutrinos and antineutrinos**. Their production rate depends only on 3 hydrodynamical variables, density (ρ), temperature (T), and chemical potential (Y_e). The energy liberated by these kinds of processes can be arranged as: $\epsilon_{brems} < \epsilon_{photo} < \epsilon_{plas} < \epsilon_{pair}$. Being the **pair-annihilation** process the only one **energetic enough** to be considered **detectable at significant distances**. Theoretically, the energy of plasmon decay process is small but not negligible, therefore it has been somewhat studied in the literature [26]. The energy spectra of these two processes are similar, but pair annihilation one depends more on temperature than

plasmon one does. Whereas, plasmon energy spectra depends more on Y_e than pair does.

Weak interaction neutrino :

- Electron Capture (EC): $(Z, A) + e^- \longrightarrow (Z - 1, A) + \nu_e$
- β^+ decay: $(Z, A) \longrightarrow (Z - 1, A) + e^+ + \nu_e$
- Positron Capture: $(Z, A) + e^+ \longrightarrow (Z + 1, A) + \overline{\nu_e}$
- β^- decay: $(Z, A) \longrightarrow (Z + 1, A) + e^- + \overline{\nu_e}$

Weak neutrino spectrum depends on temperature, density and abundances of nuclei involved. The latter is a large set of varying quantities and no simple description of it exist anywhere in the literature. Therefore, such a problem **cannot be studied** in general conditions. Particularly, abundances depend on history of the matter. However, there does exist an exception to this, the Nuclear Statistical Equilibrium (NSE). However, NSE approximation is only valid during the neutronization phase of the star and has already been studied [19]. Hence, we will not be studying it.

On the other hand, from (https://zenodo.org/records/8327401) [13], we can compute neutrino luminosity for different emission processes for a 20 $M_\odot$ star with $[Fe/H] = 0$ during its evolution. Results are shown in **Figure 2** and can be better visualized at **Appendix 2**. As we can see, thermal neutrinos surpass nuclear ones right before Carbon burning begins and continue to do so up until right before collapse.

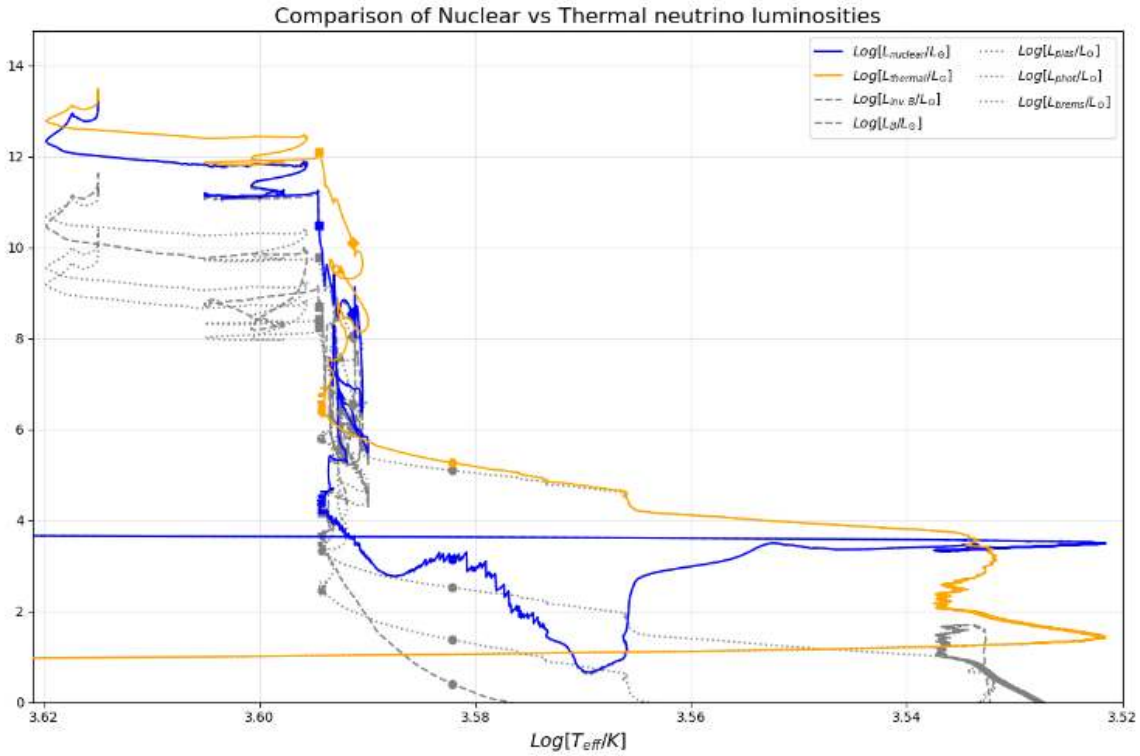


Figure 1: HR thermal and nuclear neutrino processes diagram

(Note: Nuclear luminosity (blue graph) takes into account not only WNI neutrinos but also PP and CNO ones)

3.1.2. Pair annihilation neutrino spectra

There is extensive work done on the differential rates and emissivities of the three main thermal neutrino processes. Pair-annihilation, plasmon decay and photoneutrinos. (Odrzywolek 2007; Dutta et al. 2004; Ratkovic et al. 2003; Misiaszek et al. 2006; Kato et al. 2015; Asakura et al. 2016). As we have already discussed, **pair-annihilation neutrino are the only ones that we can detect**, and hence, our main focus of interest. The Feynman diagram of this process is:



Figure 2: Feymann diagram for pair-annhilation neutrinos

The process was proposed by Chiu and Morrison (1960) and independently the same year by M.Levine (in a private communication mentioned by Fowler and Hoyle 1964). This process is most efficient in the low-density plasma of the star, where the positron fraction is the highest. Indeed, calculations show that the process is extremely efficient in a non-degenerate plasma of temperature of around $T > 10^{10}$ K. In a strongly degenerate electron plasma, however, the process is suppressed because of the negligibly small positron fraction.

As we have already discussed, unlike other neutrino processes, this one can be calculated very accurately as a **function of temperature and electron density**. The expression for the neutrino emissitivity is given by [20]:

$$Q_{\rm pair} = \frac{Q_c}{36\pi}\left\{C_+^2\left[8(\Phi_1U_2+\Phi_2U_1)-2(\Phi_{-1}U_2+\Phi_2U_{-1})+7(\Phi_0U_1+\Phi_1U_0)\right.\right.$$
$$\left.\left.+\ 5(\Phi_0U_{-1}+\Phi_{-1}U_0)\right]+9C_-^2\left[\Phi_0(U_1+U_{-1})+(\Phi_{-1}+\Phi_1)U_0\right]\right\},$$

where

$$Q_c = \frac{G_F^2}{\hbar}\left(\frac{m_e c}{\hbar}\right)^9 = 1.023 \times 10^{23}\ \text{erg cm}^{-3}\ \text{s}^{-1}$$

Is just a combination of fundamental constants and

$$f_e = \left[\exp\left(\frac{\epsilon_e - \mu_e}{k_B T}\right) + 1\right]^{-1}, \quad f'_e = \left[\exp\left(\frac{\epsilon'_e + \mu_e}{k_B T}\right) + 1\right]^{-1}$$

are the equilibrium Fermi-Dirac distributions. While the dimensionless functions U_k and ϕ_k (k= 1, 0, 1, 2) are defined by the one-dimensional integrals

$$U_k = \frac{1}{\pi^2}\int_0^\infty \frac{p_e^2\,\mathrm{d}p_e}{(m_e c)^3}\left(\frac{\epsilon_e}{m_e c^2}\right)^k f_e, \quad \Phi_k = \frac{1}{\pi^2}\int_0^\infty \frac{p_e'^2\,\mathrm{d}p'_e}{(m_e c)^3}\left(\frac{\epsilon'_e}{m_e c^2}\right)^k f'_e.$$

Explaining in detail the in and outs of the neutrino spectrum generation **exceeds the purpose** of this work. Odrzywolek [25] reproduced the spectra by appropriately integrating these formulas. Unfortunately, if one attempts to consider the detection of these neutrinos (as we do) very important **information about the neutrino energy is missing**. To **solve** this, **Monte-Carlo methods** have been used in more modern works to simulate the spectra as in [15].

On that note, since stars are not homogeneous [8], it would be necessary to calculate the number and energy emissivities per volume and time, Q_N and Q_E, as well as the spectra, $\frac{dQ_N}{dE}$, as a function of radius and integrate them over the star to obtain the number and energy luminosities, L_v^N and L_v^E together with the observed spectra, dL_v^N/dE for all flavors of neutrinos in the progenitor phase and for neutrinos other than τ_e in the collapse phase. This work has **already been done** by Odrywolek for 20 $M_\odot$ star of solar metalicity [30] .

Furthermore, as in [18] and [25], neutrino spectrum shape can be **approximated** as:

$$\phi_{\text{pair}}(\epsilon) = \frac{A}{k_B T}\left(\frac{\epsilon}{k_B T}\right)^{\gamma} \exp(-a\epsilon/k_B T).$$

Figure 3: Pair-annihilation spectrum aproximation

3.2. Neutrino Mixing, Oscillations and Hierarchies

In **quantum mechanics** [17], neutrinos can be described using **two bases**. The **flavor** one (v_e, v_τ, v_μ) and the **mass** one v_1, v_2, v_3. The flavor states are not the same as the mass states. Instead, each neutrino flavor is a **quantum mixture** (or superposition) of the mass states. This is called **neutrino mixing**.

Weak nuclear force, the force with which neutrinos interact with matter, for example, **interacts** with neutrinos via their **flavor state**. While their behavior during **propagation** in vacuum depends only on their **mass state**. The **transformation matrix** between the two basis, that is, the matrix that describes the neutrino mixing is called **PMNS** matrix and is as follows:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}.$$

It contains **angles** that determine how much the mass states overlap with the flavor states. When neutrinos travel, the **mass states evolve differently** because they have slightly different masses. This difference in evolution causes the contributions to interfere with each other, changing the mixture. When we measure v_e, (v_1, v_2, v_3) might evolve afterwards, and when we measure it again, the state might correspond to v_τ

The oscillation effects affect the neutrino flavor ratios, and hence, the rates in the detectors. The electron neutrinos and anti-neutrinos may convert to $_x$ and/or $_{\overline{x}}$, respectively, and vice versa during propagation as a result of **flavor oscillations**. In this work, we take into account both the **vacuum oscillations and MSW (Mikheyev–Smirnov–Wolfenstein)** effect that describes how neutrinos oscillates in matter and ignore the collective oscillations (density is low enough for neutrino-neutrino interactions to be negligible)

In vacuum the oscillations depend only on the differences in squared masses of neutrinos (Δm^2) the neutrino energy, (E), and the mixing angles in the PMNS matrix. That is:

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\sin^2\left(\frac{\Delta m^2 L}{4E}\right)$$

Oscillation experiments measure the **difference in squared mass, not the individual ones**. This means that we do not know the exact hierarchy of the neutrino mass states. Assuming **Normal Hierarchy (NH)** ($m1 < m2 << m3$) and **Inverse Hierarchy (IH)** ($m3 << m1 < m2$). This means that $\triangle m^2_{21} > 0$ always, but $\triangle m^2_{31}$ will be either positive or negative depending on the hierarchy.

3.2.1. Computing MSW Oscillation effect

The Mikheyev-Smirnov-Wolfenstein (MSW) effect describes how neutrino oscillations are modified when neutrinos travel **through a medium** (e.g., a star). The medium affects the oscillations because neutrinos interact differently with electrons in the medium, depending on their flavor. This interaction creates an **effective potential** that alters the oscillation probabilities.

Since there are three basis states a general neutrino wave function is described by three complex coefficients and evolves according to the Schrödinger equation. In **vacuum** the Hamiltonian is **diagonal** in the mass basis but possesses off-diagonal terms in the flavor basis that are the cause of flavor oscillations. In the presence of **matter** a **potential**, V(r), that takes into account coherent forward scattering of the neutrinos, is included in the Hamiltonian.

We are considering only neutrino flavors that have interaction with the weak force (active ones), therefore the perturbation in the Hamiltonian might be characterized as

$$V_{ee}(\mathbf{r}) \;=\; \sqrt{2}\,G_F Y_e(\mathbf{r})\,\rho(\mathbf{r})/m_N$$

The potential for antineutrinos is the same, but with opposite sign. Due to the inclusion of Vee(r) the Hamiltonian is a function of position and is neither diagonal in the mass basis nor the flavor basis. **Diagonalization is not an option** since the

unitary transformation necessary to do so also varies with position.

In general, the three complex components of the wave function oscillate simultaneously. What occurs exactly, depends upon the energy of the neutrinos E, the differences between the squares of the masses m1, m2, m3, the mixing angles that describe the relationship between the flavor and mass basis, and, of course, $V(r)$. Since there are three mass eigenstates there are **three separate mass splittings** $\delta m^2_{ij} = m^2_i m^2_j$ (though only two are independent) and the relationship between the neutrino flavor and mass bases is described by the matrix U which, in turn, is parameterized by three mixing angles θ_{12}, θ_{13} and θ_{23} plus a CP-phase δ. The structure of U is

$$U = \begin{pmatrix} c_{12}\, c_{13} & s_{12}\, c_{13} & s_{13}\, e^{-\imath\delta} \\ -s_{12}\, c_{23} - c_{12}\, s_{13}\, s_{23}\, e^{\imath\delta} & c_{12}\, c_{23} - s_{12}\, s_{13}\, s_{23}\, e^{\imath\delta} & c_{13}\, s_{23} \\ s_{12}\, s_{23} - c_{12}\, s_{13}\, c_{23}\, e^{\imath\delta} & -c_{12}\, s_{23} - s_{12}\, s_{13}\, c_{23}\, e^{\imath\delta} & c_{13}\, c_{23} \end{pmatrix}$$

where $c = cos\theta_{ij}$, and $s = sin\theta_{ij}$. Quite generally the **coherent matter** basis wave function that arrives at a distance from the core of the SN is related to the initial matter basis wave function at star via the equation $\phi_v(d, E) = S_v(d, E)\phi_v(0, E)$, where the S-matrix is the scattering or flavor time evolution matrix. An analog expression exists for antineutrinos. To determine $\psi_v(d, E)$ and $\psi_{\overline{v}}(d, E)$ we need to know both the initial states $\psi_v(0, E)$ and $\psi_{\overline{v}}(0, E)$ and the two matrices S and $S_{\overline{v}}$.

S matrices will depend on the considered medium, and the kind of neutrinos that pass through them. Normally, vacuum (V), **high density medium** (H resonance, where oscillation probability between flavors in enhanced) and **low density medium** (L resonance) are considered, the results are: $S_v = S_{Vv} S_{Hv} S_{Lv}$ and $S_{\overline{v}} = S_{V\overline{v}}\ S_{H\hat{v}}$. There is no matrix $S_{L\overline{v}}$ because anti-neutrinos do not experience L resonance in matter. Structure of this kind of matrices are present in the literature and define the crossing probabilities. On the other hand, for the initial states, depending on the hierarchies the initial matter states spectra will correspond to one flavor spectrum or another one. Calculations **have already been done** by Kneller in [17].

Putting all together, the flux at earth of certain matter state produced by a star is:

$$F_{\nu_i}(d,E) = \frac{1}{4\pi\, d^2} \sum_j |(S_\nu)_{ij}(d,E)|^2\, \Phi_{\nu_j}(0,E),$$
$$F_{\bar{\nu}_i}(d,E) = \frac{1}{4\pi\, d^2} \sum_j |(S_{\bar{\nu}})_{ij}(d,E)|^2\, \Phi_{\bar{\nu}_j}(0,E).$$

However, at earth, only flavors are detected and therefore any **coherence** that exists between matter states **is lost** and we are left with the incoherent sum:

$$F_{\nu_\alpha}(d,E) = \sum_i |U_{\alpha i}|^2\, F_{\nu_i}(d,E).$$

And implementing the neutrino survival probabilities p, and $\overline{p}$, we have that:

$$F_{\nu_e}(d,E) = \frac{1}{4\,\pi\, d^2}\ [p(E)\,\Phi_{\nu_e}(0,E) + (1-p(E))\,\Phi_{\nu_x}(0,E)]$$

$$F_{\bar{\nu}_e}(d,E) = \frac{1}{4\,\pi\, d^2}\ [\bar{p}(E)\,\Phi_{\bar{\nu}_e}(0,E) + (1-\bar{p}(E))\,\Phi_{\nu_x}(0,E)]$$

$$4F_{\nu_x}(d,E) = \frac{1}{4\,\pi\, d^2}\ [(1-p(E))\,\Phi_{\nu_e}(0,E) + (1-\bar{p}(E))\,\Phi_{\bar{\nu}_e}(0,E) + (2+p(E)+\bar{p}(E))\,\Phi_{\nu_x}(0,E)]$$

where Dighe-Smirnov [38] notation has been adopted (υ_μ, υ_τ, $\overline{\upsilon_\mu}$, $\overline{\upsilon_\mu} = \upsilon_x$) and p are the survival probabilities of the neutrinos, that is [17]:

The probabilities, are given by the adiabatic limit (crossing probability close to 0) [10]. And their values are set [19] as $p = 0.02/0.3$ for NH/IH and $p = 0.02/0.3$

$$p(E) = \sum_i |U_{ei}|^2 \, |(S_\nu)_{ij}(E)|^2$$
$$\bar{p}(E) = \sum_i |U_{ei}|^2 \, |(S_{\bar{\nu}})_{ik}(E)|^2$$

3.3. Detectability

As we know, pair-annihilation neutrinos produce all kinds of flavors of neutrinos and anti-neutrinos. Different methods of detection exist depending on which kind of flavor we are dealing with as well as if the particle in question is a neutrino or an anti-neutrino. Specifically we will study the following methods: **neutrino electron scattering (ES)**, **inverse beta decay (IBD)**. For simplicity, the equation we will be using to estimate the rates is:

$$R = \overline{\sigma} \cdot N \cdot \phi_\alpha \cdot \epsilon \cdot$$

Where, $\overline{\sigma}$ is the average cross section of the neutrinos. N is the number of targets in the detector ϕ_α is the number flux of pertinent neutrinos and ϵ is the efficiency of the detector. The aforementioned methods will determine the pertinent cross sections as well as the kind of targets. While the specific detectors considered will determine the number of targets and the efficiency of the process.

3.3.1. Neutrino-electron scattering

This process [3] describes how a **neutrino interacts with an electron via the weak force**. It produces recoil electrons with a kinetamic energy that goes from zero up to the kinetic maximum. On the other hand, the weak interaction's chiral nature significantly affects the scattering probabilities, making the process sensitive to the neutrino's flavor and helicity.

There are two main interaction types for neutrino-electron scattering, depending on the flavor of the neutrino and the nature of the interaction.

Charged Current (CC): Charged current occurs when the weak interaction between the electron and the neutrino is mediated by a **charged W boson**. This process

only affects electronic neutrinos.

$$v_e + e^- \longrightarrow e^- + v_e$$

Neutral Current (NC): When the interaction in mediated by a **neutral Z boson**. This process affects all flavors of neutrino.

$$v_x + e^- \longrightarrow v_x + e^-$$

The general formula for the **cross section** of ES is as follows [30]:

$$\frac{d\sigma}{dT} = \frac{2G_F^2 m_e}{\pi}\left[c_1^2 + c_2^2\left(1 - \frac{T}{E_\nu}\right)^2 - c_1 c_2 \frac{m_e T}{E_\nu^2}\right]$$

where: c_1, c_2 are the effective coupling constants that depend on the type of interactions, G_F is the fermi constant, T is the kinetic energy of the scattered electron and E_v is the initial energy of the neutrino. According to [39] we have the different coefficients for the different flavors of neutrino.

Reaction	c_L	c_R	$c_L^2 + \frac{1}{3}c_R^2$	$\frac{1}{2}c_L c_R$
$\nu_e e^-$	$1/2 + sin^2\theta_W$	$sin^2\theta_W$	0.5525	0.0845
$\bar{\nu}_e e^-$	$sin^2\theta_W$	$1/2 + sin^2\theta_W$	0.2317	0.0845
$\nu_{\mu,\tau} e^-$	$-1/2 + sin^2\theta_W$	$sin^2\theta_W$	0.0901	-0.0311
$\bar{\nu}_{\mu,\tau} e^-$	$sin^2\theta_W$	$-1/2 + sin^2\theta_W$	0.0775	-0.0311

Figure 4: ES coefficients [30]

As we can see, cross section for μ and τ neutrinos/anti-neutrinos will be the same. Although ES cross sections are low, it has **no energy threshold** which makes it viable to study neutrinos. Moreover, since the scattered electron's trajectory is correlated with the incoming neutrino's direction. What makes ES really valuable is the **directional information** it provides.

3.3.2. Inverse Beta Decay

Neutrino-proton scattering is often referred to as IBD or inverse beta decay. With this process, we are able to study only electronic anti-neutrinos, however, compared to ES, it has higher cross sections for higher energies and less background. When possible, it becomes the dominant channel for slightly more energetic $\overline{v_e}$ detection. This process is a **charged current one**, this is CC.

$$\overline{v_e} + p \longrightarrow e^+ + n$$

The low background is due to the **two signals** that this process emits.

Positron signal: The e^+ carries the majority of the energy of the original anti-neutrinos. The positron **quickly** annihilates with an electron in the detector medium, producing two gamma rays: $e^+e^- = 2\gamma$ which produces the promt signal in the detector.

Neutron signal: The neutron (n) produced in the reaction is initially 'free' and is eventually captured by a nearby nucleus liberating a gamma ray. The energy is usually 2.2MeV is hydrogen capture is used. Or 8MeV for capture in gadolinium. This signal is recieved some mili/micro seconds **after the positron signal**.

The process of corroborating the coincidence between the positron and delayed neutron signal is called **Delayed Coincidence** (DC). And it is what makes IBD, in general more potent that ES since with it, as mentioned before, we are able to **discard backgroud noise** and to confirm true IBD events. The Cross-section for IBD is given by [30]:

$$\sigma = 0.0952 \ (\frac{E_e p_e}{1 MeV^2}) \cdot 10^{42}$$

Where

$$p_e = \sqrt{E_e^2 - m_e^2}$$

The drawbacks of this method is that it provides little to **no directional information** and has an energy threshold of 1.8 MeV. Yet, it is still better than ES for high energy. For lower energies however, the lower cross sections and the threshold puts IBD a little bit behind ES. But since our neutrinos follow a certain energy spectrum, ES

would be better than IBD for certain regime and vice-versa, overall, which method is better between IBD and ES for our purposes is **not clear** whatsoever.

3.3.3. Other methods of detection

There are other detections methods that we will not study but that are, still, worth mentioning. The first of those is **Argon absorption**.

$$v_x + {}^{40}Ar \longrightarrow e^- + {}^{40}K^*$$

This process is exclusively for neutrinos. The majority of the more interesting events would be produced by CC interaction which is for electronic neutrinos only. NC would have less impact, but with it we would be able to detect all flavors of neutrinos. Theoretically, it is a possible way to detect pair-annihilation neutrinos from a neutrino-cooled star. However, its relatively **high threshold** for our purposes (5 MeV) makes it almost completely **unusable**. The SURF DUNE detector, with its 4 LArTPC (Liquid Argon Time Projection Chambers) is the most advanced Argon detector in the world. This method of detection cannot 'see' further back than the inmediate pre-supernova stage, and hence, we will not be studying it.

Another important detection method are the **interactions with heavy water**.

$$v_x + d \longrightarrow e^+ + n + n$$

This process is suitable for all kinds of neutrinos and antineutrinos. CC channel is again only for electronic neutrinos and antineutrinos and NC is for all of them. Only difference to take into account is that the threshold for neutrinos is 2.2 MeV whereas the one for antineutrinos is 4 MeV. This detection method has mainly been utilized by the SNO detector (now inactive). With it, we would be able to see the pair-annihilation neutrinos we are interested to see. However, the high energy threshold (2.2 - 4 MeV) and the low cross sections summed with the considerable background, make it **significantly worse** that the two main options already discussed.

3.4. Detectors

When it comes to neutrino detectability, there are different technologies on which the different detectors are based.

3.4.1. Water Cherenkov

Water Cherenkov detectors detect **Cherenkov radiation**, which is light emitted when a **charged particle** travels through a medium **faster than the speed of light** in that medium. This phenomenon is analogous to a **sonic boom**, but with light instead of sound

Although relatively cheap and scalable [24], they do not have very good efficiencies with respect to ES events below an energy threshold of 5 MeV. Therefore, the **majority** of events, will be produced by the **NC/CC** interactions of the IBD channel.

In the more advanced detectors, **gadolinium** is introduced into the water to further improve the neutrino detection. Gadolinium is a chemical element with a high neutron capture capacity. This **improves the efficiency** of ES and the detection of the delayed signal of the IBD

3.4.2. Liquid Scintillator

Liquid scintillator detectors work by using a liquid that emits light (scintillation) when it absorbs the energy from a passing charged particle. Scintillation is the emission of flashes of light when certain materials are excited by ionizing radiation.

They **are bette**r than Water Cherekov ones for **IBD**, since they have a more efficient positron + neutron capture efficiency. Regarding ES, LS detectors are very sensitive to low energy depositions, however the **efficiencies** are slightly **worse** than WC ones [23]. They also can be doped with gadolinium.

3.4.3. Detectors considered

In this work, we will consider **3 different detectors** as subjects for the pair-annihilation neutrino rates. These are **KamLAND** and **JUNO**, both liquid scintillators. And **SK-GD**, a water cherenkov detector. [10] [22] [23] [24] [34]

Detector	**KamLAND**	**JUNO**	**SK-Gd**
Type	LS	LS	WC
Size	0.7kt	32.5	20
N_p	$8.55 \cdot 10^3 1$	$1.69 \cdot 10^3 3$	$2.14 \cdot 10^3 3$
N_e	$3.43 \cdot 10^3 2$	$9.3 \cdot 10^3 3$	$1.67 \cdot 10^3 4$
ϵ_{ES}	55%	65%	45%
ϵ_{IBD}	75%	73%	52%

Table 1: Characteristics of detectors used.

For the backgrounds it is necessary to bear in mind the type of detector, the specific channel of detection studied and the specific characteristics of the detector itself. Unlike IBD events, which can be identified by coincidence, ES causes single hits in the detector and suffers from a higher background. The table below summarizes all the backgrounds we should take into consideration for our energy range of (0,5) MeV.

Source	**KamLAND**	**JUNO**	**SK-Gd**
Solar neutrino (pp, B^8, pep)	10	200	75
Reactor	10	20	Negligible
DSNB	Negligible	Negligible	Negligible
Cosmogenic ^{11}C	35	220	Negligible
Spallation Neutrons	Minor (1 to 2)	Minor (1 to 5)	Taggable
Other (Radiation, Geo-ve,external gamma)	Few (1 to 5)	Few (5 to 10)	Negligible
Total	-57	-445	75

Table 2: Expected background events for electron-neutrino scattering in each detector.

For ES, atmospheric neutrino are not of our concern below 10 MeV, B^8 solar neutrinos are the only ones which are **non-negligible**. DSNB (Diffuse supernova neutrino background) is important above 10 MeV but is also negligible below this energy. Reactor neutrinos produce $\overline{v}_e$ that are important in our energy range. Cosmogenic ^{11}C rates are high, but **mitigation strategies** (veto and tagging techniques about muon trunks) are adopted leaving a manageable 3 to 4 events a day for KamLAND and 20 to 24 for JUNO [40]. For SK-GD, since it is a Water Cherenkov detector and uses water instead of carbon as target material, comosgenic ^{11}C is not of our concern. (^{11}C background is produced when cosmic ray muons interact with carbon nuclei in the LS through spallation reactions $\mu +^1 2C \longleftrightarrow^1 1C + n + other$). Lastly, spallation neutrinos

(neutrons produced by gamma rays) are of the order of 1 to 5 BG events per day on all detectors.

On the other hand, IBD events usually suffer from a lower background due to the events being **taggable** via DC. Usually, reactor neutrinos are of concern, and, account for all the $\overline{v}_e$ sources mentioned before. We also must consider more seriously other background sources that were not considered before such as geo_{v_e} rates. [10] Gives:

Source	**KamLAND**	**JUNO**	**SK-Gd**
Total	0.7	20	24

Table 3: Expected background events for electron-neutrino scattering in each detector.

4. Methodology

To simulate stellar evolutions, we use the simulations already done by **Ebraheem Farag** using **MESA** version r15140. Details of his calculations are presented in [13]. Out of the 120x5 paths (120 solar mass x 5 different metallicites) we use the 20 solar mass one but for the most realistic metallicities of $[Fe/H] = 0.5, 0, -0.5$

The code gives as output, separately, the pair-annihilation, plasmon, photon, and bremastahlung luminosity in (erg/s). Only pair annihilation is studied, dividing the array between all the burning stages according to [13]. To get the corresponding number luminosities, the pair luminosities are divided by the average energy of the neutrinos given by Odrzywolek [30]. Then, the number pair annihilation luminosities for each neutrino flavor for each burning stage is obtained by **computing the ratios**, again, given by Odrzywolek [30].

Subsequently, multiplying by the geometric factor the luminosities and taking into account the ratios for each flavor as well as the vacuum and MSW oscillation effects already discussed. **Number luminosities** on earth for each flavor and for each burning phase are calculated.

Burning phase	ν_e ($\bar{\nu}_e$) fraction	$\nu_{\mu,\tau}/\nu_e$ ratio	Average ν_x energy
C	42.5 %	1:11.4	0.71 MeV
Ne	39.8 %	1:7.8	0.99 MeV
O	38.9 %	1:6.9	1.13 MeV
Si	36.3 %	1:5.4	1.85 MeV

Figure 5: Flavor ratios [30]

On the other hand, the pair neutrino spectrum (which is the same for all flavors of neutrinos) is obtained by Odrywolek following the **Monte-Carlo** method of [41] , which is derived from the one developed by Janka in [15].

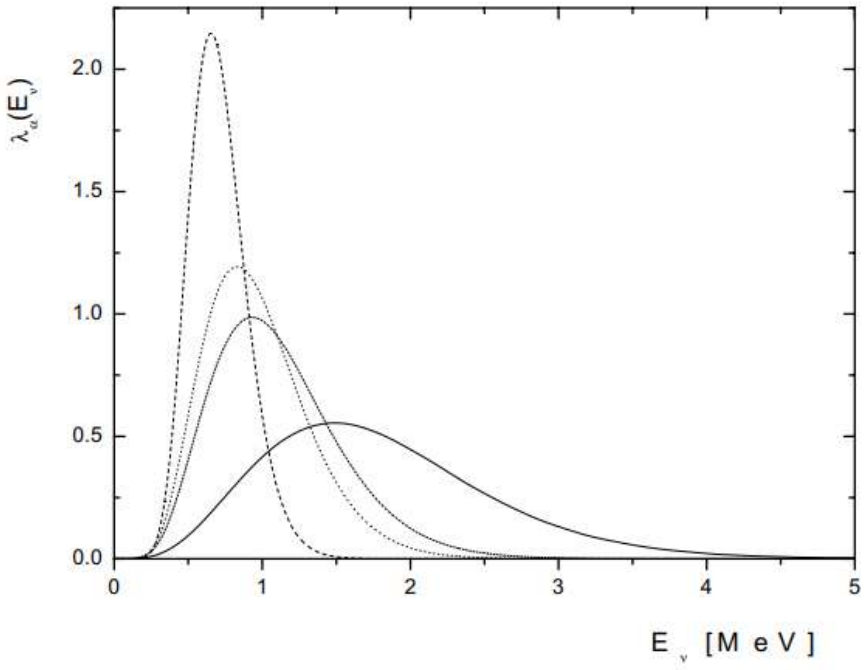


Figure 6: Normalized spectra [30]

Using this spectrum, the average cross section of the neutrinos is calculated by using:

$$\hat{\sigma} = \int_0^E \sigma(E)\lambda(E)dE$$

Where $\sigma(E)$is calculated using the formulas either for ES or IBD presented before and $\lambda(E)$ is the normalized pair spectra.

Rates on different detectors are calculated using the aforementioned **rate formula** and following the specific characteristics presented at **Table 1**. Cumulative events for each burning phase can be calculated integrating rates over the time. Moreover, BG events are computed assuming them as **constant**. The analysis is then made for all the different flavors using either **ES** channel either **IBD** one when pertinent and taking into account **both of the neutrino mass hierarchies** [13]. This process is done for all **3 different metallicities** and **different distances** of interest.

5. Results

Results are **presented** for a 20 solar mass star of solar metallicity situated at **0.2 kpc** of distance (such as **Betelgeuse**). For the IBD channel, we would get:

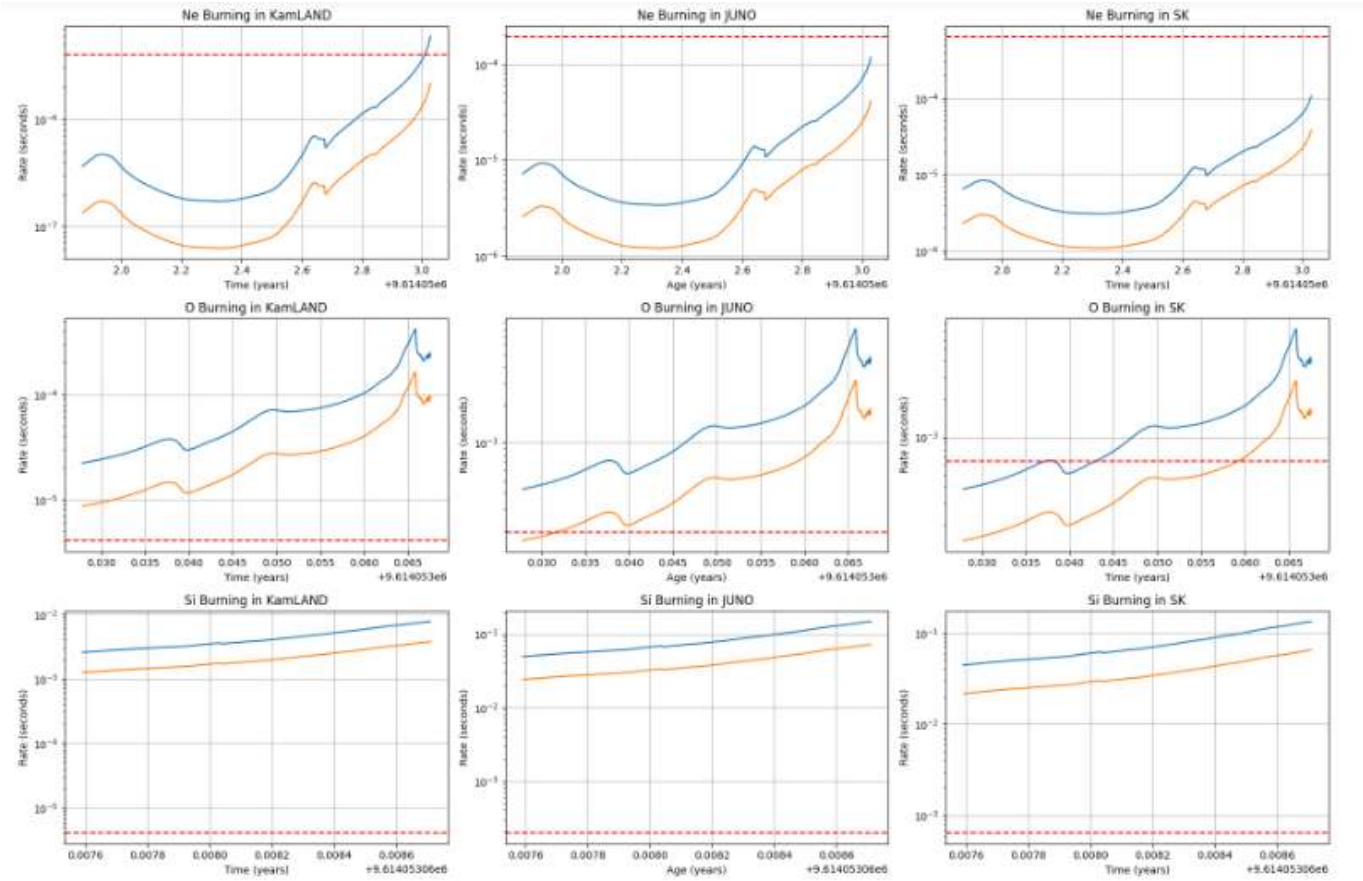


Figure 7: IBD rates for KamLAND, JUNO and SK-GD during different burning stages. NH is represented in blue. IH in orange.

For ES, we would have, assuming NH:

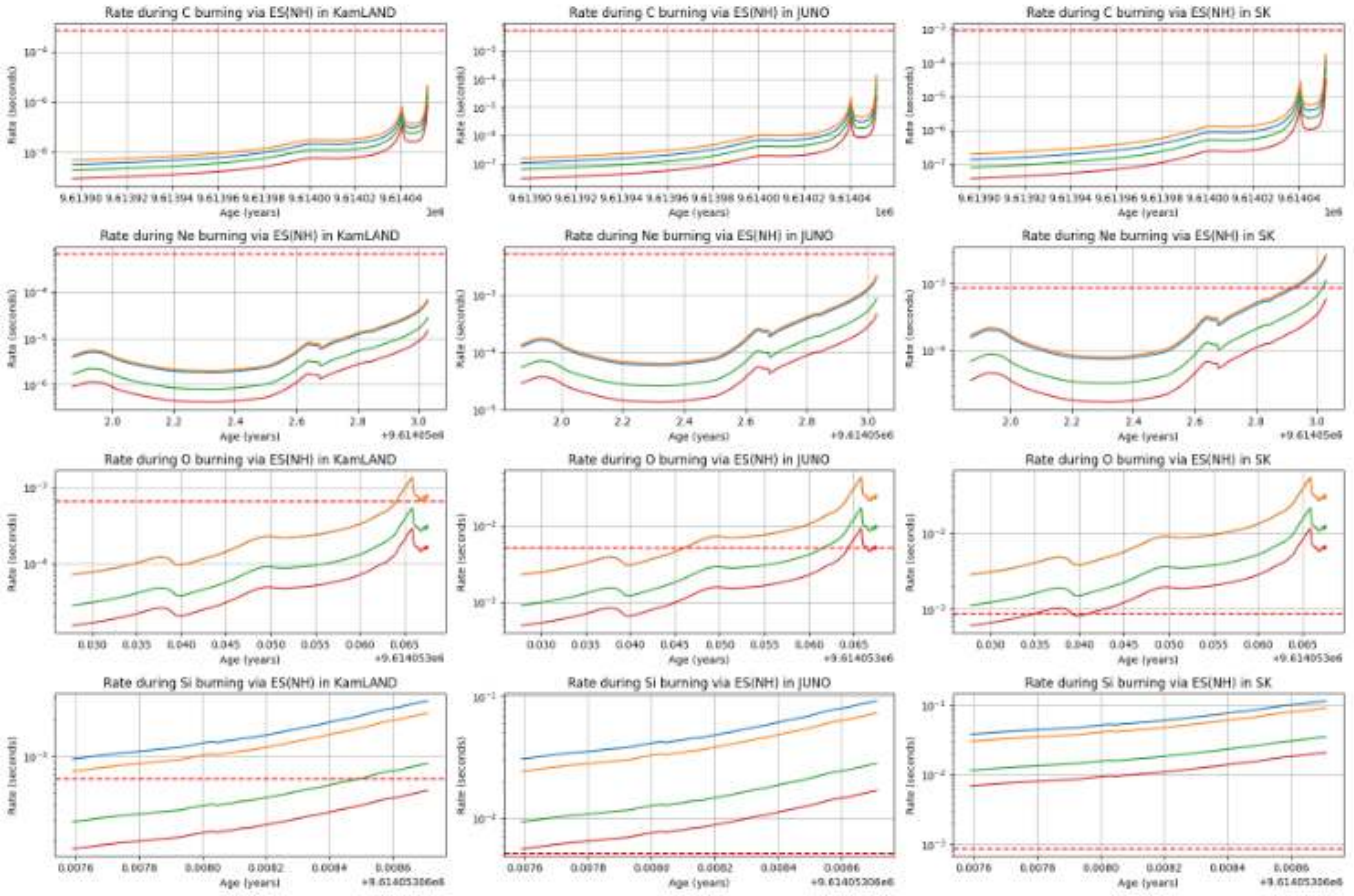


Figure 8: ES NH Rates for KamLAND, JUNO and SK-GD during different burning stages. Blue is electronic antineutrinos, Orange is electronic neutrinos, green is mu/tau neutrinos and purple is mu/tau antineutrinos.

And assuming IH:

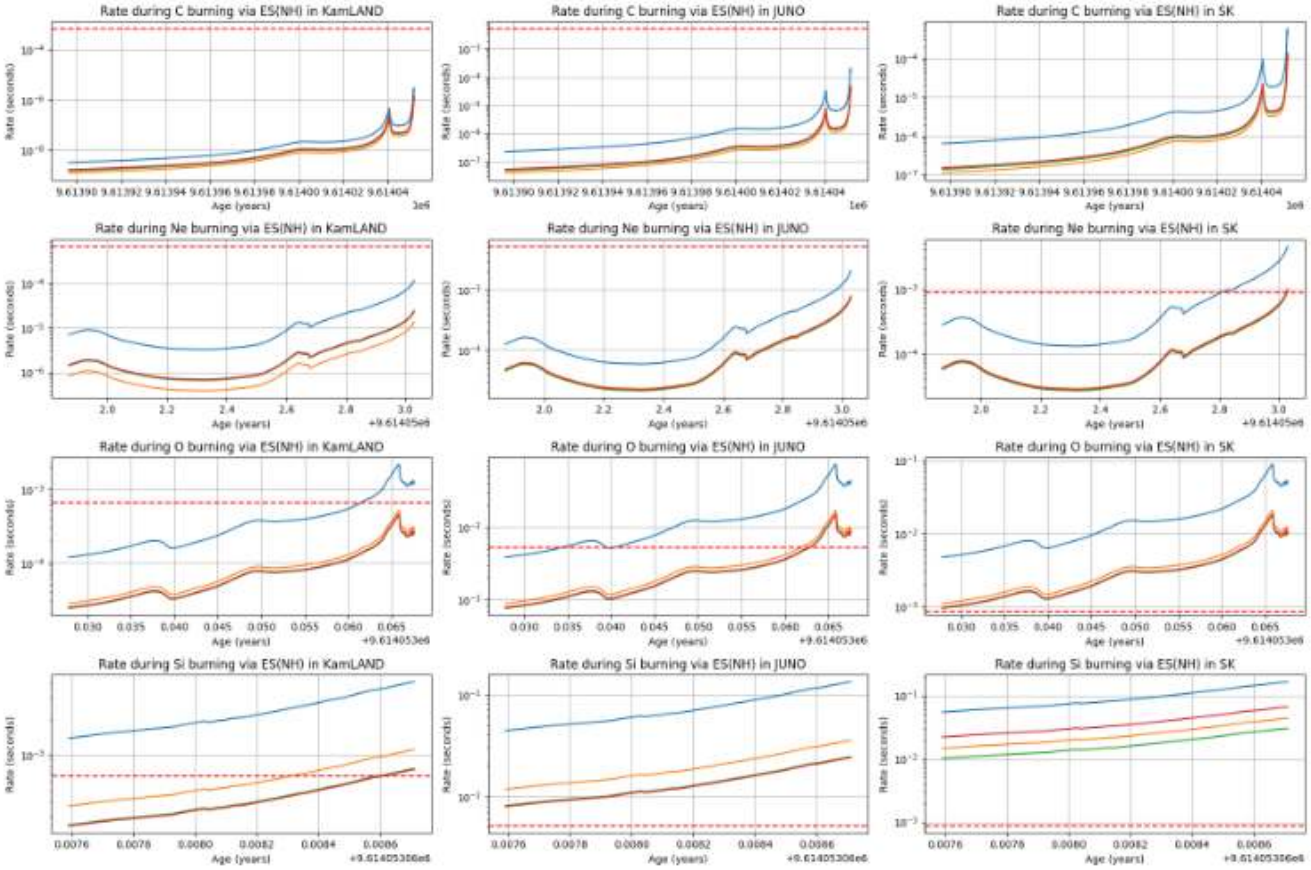


Figure 9: ES IH Rates for KamLAND, JUNO and SK-GD during different burning stages. Blue is electronic antineutrinos, Orange is electronic neutrinos, green is μ/τ neutrinos and purple is μ/τ antineutrinos

As we can see, $\overline{\nu}_e$ are the **first ones** to be detected for both hierarchies and for both, ES and of course IBD. Centering the analysis on the NH diagram. Without computing the thresholds, IBD only **surpasses** ES in total cumulative events **during Si burning** due to higher cross sections at higher energies. However, once computed the BG events, **IBD is the most suitable** method of detection **for LS** detectors due to lower background. For our Water Cherenkov detector (SK-GD), nonetheless, the situation is quite different. **SK-GD** has **better results for ES than for IBD** and it is by far the **best detector for ES while being the worst for IBD**. This is due to its low background compared with its larger fiducial paired with a still low but improved sensitivity after Gd doping.

6. Conclusions

Out of the 2 **LS** detectors,**KamLAND** has the **best results for IBD** while **JUNO** has the **best ones for ES**. Yet, since IBD is far better for LS than ES, we **would choose KamLAND** over JUNO to study a neutrino-cooled star. Comparing between KamLAND IBD results vs SK-GD ES results, one finds that with SK-GD, we would be able to study the neutrino-cooled star a little bit before than with KamLAND (only by a factor of a few days). Hence, making SK-GD the go-to detector during the neutrino-cooled stage. However, it must be pointed out that **efficiencies and backgrounds for ES overall, are somewhat optimistic compared to IBD** which favors SK-GD in detriment of KamLAND and JUNO.

In reality, if we were to assume more realistic efficiencies and backgrounds for SK-GD, **KamLAND** would rise as the best detector which makes it the most reasonable **go-to detector in order to best analyze our star**. Now, taking KamLAND into consideration, $\overline{\nu}_e$ are the first neutrinos we would observe. At ~ 0.3 kpc we will only see **O burning onwards**. At $\sim$ 2kpc only **Si onwards**, and, finally, at ~ 10 kpc we will **only see the collpase-phase**. This is, only the supernova. This results, changes a bit with metallicity. In principle, the lower the **metallicity** the sooner we will be a able to study the star. This may be due to higher temperatures and densities in the core that directly affect pair annihilation production. However, present-day metallicity differences in the Galactic disk are smaller than the range of metallicites studied here.

Potentially, assuming **NH** and again taking into consideration **KamLAND** results, we could analyze via neutrinos our 20 solar mass star **5 months and half before collapse**, if the models are truly quantitative in their predictive time scale. For **IH**, the discussion is pretty much the same. ν_e-antineutrinos are still the ones we would detect first, but for IH they would arrive a little bit before than for NH In total, considering IH, the effective time exposure of KamLAND to the star can be expanded to **almost 6 month**s. This result can help improve the supernova early warning systems (SNEWS) already in place.

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8. Apendix 1

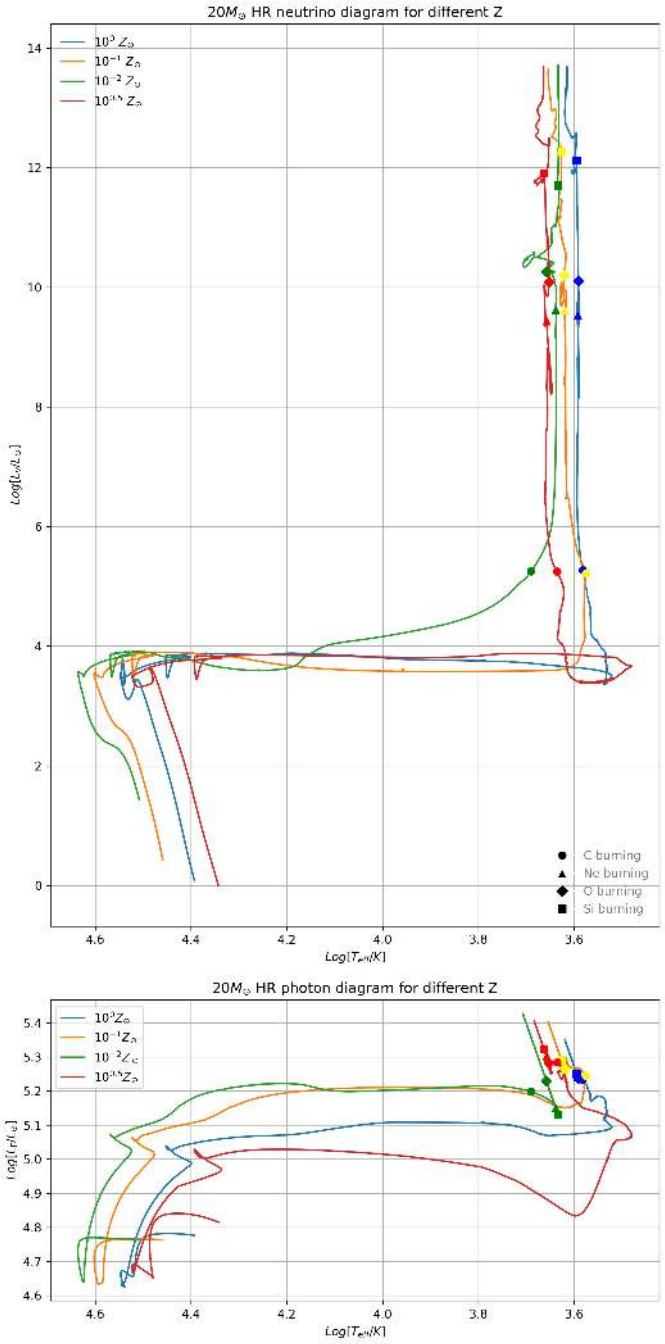


Figure 10: Schematic view of the neutrino emission of 20M0 massive star (Odrzywolek)

9. Apendix 2

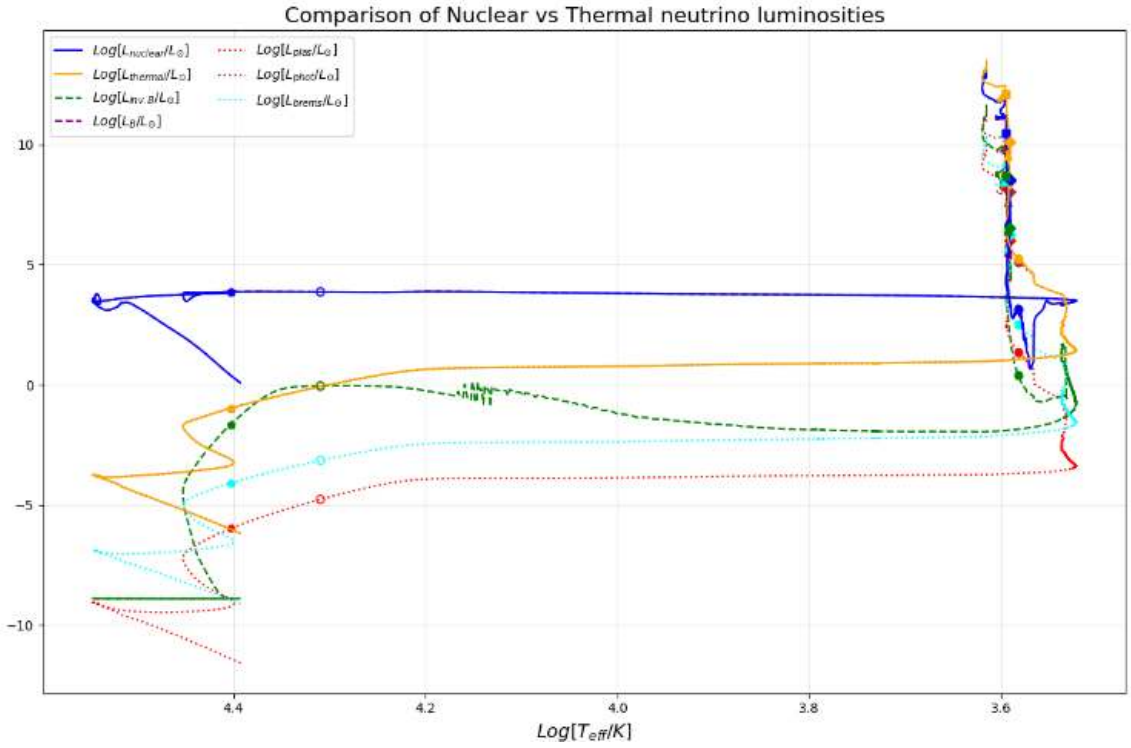


Figure 11: HR thermal and nuclear neutrino processes diagram

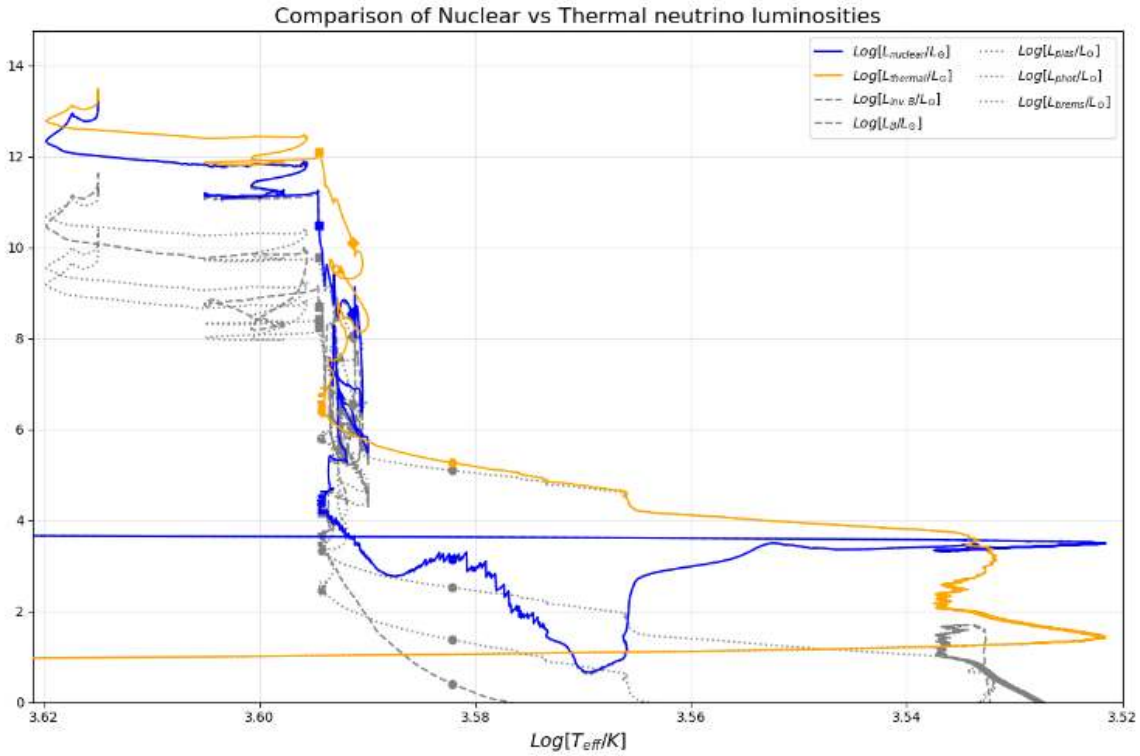


Figure 12: HR thermal and nuclear neutrino processes diagram (Simplified)